The Application Gap: Why Small Counties Receive Fewer Federal Disaster Mitigation Grants

Research Article

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Abstract

Federal grants can help communities reduce the damage from future disasters, but receiving these funds often depends on whether local governments apply. This paper examines whether smaller counties are less likely to access such funding through the Hazard Mitigation Grant Program (HMGP), a major source of federal disaster-mitigation funding in the United States. We examine 12,081 county–disaster declaration observations covering 2,767 counties and 736 tornado and severe-storm declarations between 1999 and 2021. Using records of both successful and unsuccessful applications, we compare counties included in the same disaster declaration. We find that the smallest third of counties are 8.8 percentage points less likely to receive an HMGP project than larger counties, whose funding rate is 27.1%. Small counties are also about half as likely to receive funding for tornado safe rooms or wind retrofits. This difference arises mainly because smaller counties apply less often, not because their applications are more likely to be rejected. Having fewer eligible local governments accounts for about two-fifths of the gap, but even county governments themselves are less likely to apply in smaller counties. Differences in local fiscal resources do not explain the remaining gap, and a federal reform reimbursing grant-management costs did not measurably reduce it. However, the gap is smaller in states where state agencies submit a larger share of applications. These findings show that disparities in access to federal mitigation funding emerge well before funding decisions are made. Improving access to the application process may therefore be important for reducing geographic inequalities in public investment.

Keywords: intergovernmental grants; hazard mitigation; local government capacity; rural counties; regional disparities

JEL codes: H77, H84, R50, Q54

1. Introduction

When a tornado or severe storm leads to a presidential disaster declaration, the federal government makes money available not only for recovery but also for reducing the damage from the next event. The main channel is the Federal Emergency Management Agency's (FEMA) Hazard Mitigation Grant Program (HMGP). Its funding is triggered by the declaration and sized as a share of the federal assistance that the disaster generates (Horn 2022). Between 1996 and 2019 the program awarded $22.4 billion, far more than FEMA's pre-disaster mitigation programs (Horn 2022). Studies of FEMA mitigation spending report benefits well above costs (Rose et al. 2007; Godschalk et al. 2009) and measurable reductions in later losses (Davlasheridze, Fisher-Vanden, and Klaiber 2017; Miao and Davlasheridze 2025). Where this money is spent therefore affects which places carry less residual risk, and lower expected losses, into the next storm.

The money does not flow automatically to the places a disaster struck. HMGP is an intergovernmental grant with a state intermediary: funds go to the state, which selects among subapplications prepared by counties, cities, school districts, special districts, and other eligible entities (Horn 2022). Each subapplicant must hold a FEMA-approved local mitigation plan, show that the project is cost-effective, and usually provide a 25 percent nonfederal match (Horn 2022; Frimpong, Reilly, and Niemeier 2023). A declaration thus creates an opportunity for federally financed public investment, but the opportunity becomes a project only if a local government decides to apply, scopes a project, and prepares the documentation. Counties that share the same disaster declaration can therefore end up with very different levels of mitigation investment. Political incentives already favor spending on relief over spending on preparedness (Healy and Malhotra 2009), and the distributional consequences of FEMA's risk-reduction programs are a recurring concern (Noonan and Sadiq 2018; Miao, Davlasheridze, and Reilly 2025).

Small jurisdictions have reasons to apply less often. Much of the work of preparing an application does not shrink with the size of the place: an engineering estimate, a benefit–cost analysis, environmental review, and the administration of a grant require similar specialized effort

whether the project serves a county of a few thousand residents or one of several hundred thousand. When these costs are largely fixed, a small government pays more per resident to obtain the same type of project, and it is less likely to employ staff who prepare such applications regularly. Research on intergovernmental grants finds that the localities with the greatest needs are often the least equipped to compete for funds (Collins and Gerber 2006; Hall 2008). In hazard mitigation, rural local governments with lower measured capacity secure less competitive federal resilience funding (Smith et al. 2025), FEMA programs that are administered through state and local governments tend to favor better-resourced counties (Miao, Davlasheridze, and Reilly 2025), property buyouts have concentrated in wealthier and more populous counties (Mach et al. 2019), and rural counties have received less HMGP assistance per recipient than urban counties (Seong, Losey, and Gu 2022). This fixed-cost reasoning motivates the analysis. Our design does not measure application costs directly, and we do not treat the reasoning as a tested result.

Three features of the existing evidence limit what it can say about small places. First, most studies observe only awards, so a shortfall in funding cannot be attributed to fewer applications, more rejections, or lower need. Second, comparisons across counties hit by different disasters, in different states and years, mix differences in local governments with differences in hazard severity, state administration, and the size of the HMGP allocation. Third, wind hazards have received less attention than floods, although tornado safe rooms and wind retrofits are a large share of HMGP construction and have long been studied for their cost-effectiveness (Simmons and Sutter 2006).

This article asks whether, and at which stage, small counties fall behind in turning tornado and severe-storm disaster declarations into funded mitigation projects. We assemble every county designated in a tornado or severe-storm major disaster declaration from 1999 to 2021 and link each county–declaration to the HMGP subapplications and projects filed under that declaration, including requests that were later withdrawn or denied. Tornado exposure is measured from Storm Prediction Center (SPC) tracks that crossed populated block groups, so that exposure does not depend on damage reports. Counts and finances of local governments come from the Census of Governments. The central comparison is between counties designated in the same declaration,

which share the incident, the state administration, and the HMGP allocation.

Within the same declaration, the smallest third of counties are 8.8 percentage points less likely to obtain any HMGP project, against a mean of 27.1 percent among larger counties, and about half as likely to obtain a safe room or wind retrofit. Participation falls steadily with population across the whole distribution rather than only among the very smallest counties. The shortfall arises before any request is reviewed: conditional on applying, small counties have their requests approved, and withdraw them, at rates similar to those of larger counties. About two-fifths of the gap is associated with the smaller number of municipalities, school districts, and special districts that could apply in small counties. A gap remains for the county government itself, of which every county has exactly one. When the same county is designated in several declarations, a tornado that crosses a populated place is followed by more applications in larger counties, whereas the estimated response of small counties is negative and imprecise; the difference between the two is marginally significant. Per-resident fiscal measures do not account for the gap, a 2018 federal provision that reimburses subrecipients' management costs did not detectably narrow it, and the gap is smaller in states whose agencies historically filed a larger share of applications.

The article contributes to the regional economics of intergovernmental transfers in three ways. First, it locates a geographic disparity in access to federally funded public investment at a specific administrative margin, the decision to apply, using application records that include withdrawn and denied requests. Second, it separates a component of the county-size gap that reflects the institutional structure of local government, namely the number of potential applicants, from a residual that remains for a single, comparable applicant in every county. Third, it examines several explanations proposed in the literature, including fiscal constraints, state delivery of applications, and federal reimbursement of administrative costs, within a common comparison. The evidence is observational: it shows where and for which places the gap arises, not the causal effect of local administrative capacity. For policy, the results suggest that disparities in grant-funded investment across places can arise even when funding formulas respond to hazards, because access depends on whether a local government applies.

Section 2 describes the program and related research. Section 3 presents the data and empirical approach, Section 4 the results, and Section 5 their interpretation, policy implications, and section 6 concludes.

2. Institutional Background and Related Literature

2.1 HMGP as an Intergovernmental Grant

HMGP is authorized by Section 404 of the Stafford Act and becomes available in a state after a major disaster declaration. The amount available for a declaration is set on a sliding scale, up to 15 percent of the first $2 billion of estimated federal disaster assistance, with higher percentages for states with enhanced mitigation plans (Horn 2022). Although the declaration designates particular counties for disaster assistance, the state manages the mitigation funds as the recipient and decides which subapplications to fund (Horn 2022). Eligible subapplicants include state agencies, local governments, and certain nonprofits; households benefit only through a community application. HMGP accounted for about 80 percent of FEMA's Hazard Mitigation Assistance spending between 1989 and 2019 (Frimpong, Reilly, and Niemeier 2023).

The program has three features that matter for the distribution of funds across places. The pool of money is set by the declaration, so counties within a declaration draw on the same allocation under the same state administration. Funds are matched, so a subapplicant must find the nonfederal share. And money reaches a place only through a subapplication from an eligible entity located there, so the number and activity of local governments determine how many claims a county can make on the allocation.

For tornado and wind hazards, HMGP finances community and school safe rooms, individual residential safe rooms through state-run programs, and wind retrofits of public and private structures. Tornado fatalities are concentrated among residents of mobile homes and in the Southeast (Ashley 2007; Sutter and Simmons 2010), the cost-effectiveness of shelters depends on housing type and local risk (Merrell, Simmons, and Sutter 2005; Simmons and Sutter 2006), and the geography of tornado activity has been shifting toward the Midwest and Southeast (Gensini and Brooks 2018).

FEMA has twice tried to reduce the administrative burden of such projects. Pre-calculated benefits allow residential safe rooms that meet design criteria to forgo a project-specific benefit–cost analysis (Federal Emergency Management Agency 2015, 2024). Section 1215 of the Disaster Recovery Reform Act of 2018 (DRRA) (U.S. Congress 2018) allows subrecipients to recover management costs of up to 5 percent of their subaward, a provision that FEMA applies to disasters declared on or after August 1, 2017 (Federal Emergency Management Agency 2023). We use the second reform as a test in Section 4.4.

2.2 Local Government Size, Capacity, and Access to Federal Funds

Research on how individuals and organizations respond to hazards emphasizes that action depends on whether a response is judged feasible and within available resources, not only on perceived risk (Lindell and Perry 2012; Bubeck, Botzen, and Aerts 2012). For a local government, the relevant resource is administrative capacity: staff time, technical expertise, fiscal room for a cost share, and experience with federal programs. Planning research shows that the involvement of professional planners changes the content of local mitigation plans (Lyles, Berke, and Smith 2014) and that federal and state policies shape local mitigation choices through local commitment and capacity (Burby 2006; Berke, Lyles, and Smith 2014). Local governments learn from their own hazard history, faster under some conditions than others (Brody et al. 2009). Organizational capacity predicts local flood mitigation (Brody, Kang, and Bernhardt 2010), and participation in FEMA's Community Rating System rises with local capacity and tax base (Posey 2009; Sadiq and Noonan 2015; Li and Landry 2018); interviews with program coordinators and floodplain managers describe why communities do or do not take part (Sadiq, Tyler, and Noonan 2020).

A second body of work studies how federal disaster resources are distributed across places. It builds on evidence that the capacity to prepare for and recover from hazards is unevenly distributed geographically (Cutter, Boruff, and Shirley 2003; Cutter and Finch 2008) and that rural and urban counties differ in measured resilience (Cutter, Ash, and Emrich 2016). Studies of household assistance report distributive inequities in how aid relates to social vulnerability (Drakes et al. 2021; Emrich, Aksha, and Zhou 2022), and programs that require government applicants tend to favor

counties in better socioeconomic condition (Miao, Davlasheridze, and Reilly 2025). Buyouts show disparities by county wealth and race (Mach et al. 2019; Elliott, Brown, and Loughran 2020), and rural HMGP recipients receive less assistance per recipient than urban ones (Seong, Losey, and Gu 2022). In competitive pre-disaster programs, local capacity predicts success (Smith et al. 2025), and small and impoverished communities obtained 29 percent of the funding they requested in the first round of FEMA's Building Resilient Infrastructure and Communities program (Smith, n.d.). The allocation of declarations themselves has a political dimension (Garrett and Sobel 2003; Reeves 2011), which is one reason we compare counties within a declaration.

States stand between the federal program and local applicants. A national survey of state hazard mitigation officers found large differences in states' ability to help localities develop applications and little use of funding for pre-application work (Smith and Vila 2020; Vilá et al. 2022). Research on other grant programs finds that the state's design of the application process can widen or narrow disparities among localities (Collins and Gerber 2006, 2008).

This work has not established where in the grant process small jurisdictions fall behind, or how much of any gap reflects the structure of local government rather than the behavior of individual governments. A county with few incorporated places, school districts, and special districts has fewer entities that could apply, so it would obtain fewer projects even if each entity were as likely to apply as its counterparts elsewhere. Separating this structural channel from a constraint on individual governments requires application-level data and a comparison that holds the hazard and the administering state fixed.

2.3 Expectations

We organize the analysis around four expectations. (H1) Conditional on hazard exposure, damage, and the declaration itself, small counties are less likely to obtain HMGP projects. (H2) If the gap reflects local conditions rather than federal or state screening, it arises at the application stage rather than in decisions on submitted requests. (H3) If the gap is not solely a consequence of the number of eligible entities, it persists after accounting for that number and appears for the county government itself. (H4) If local capacity constrains the response to hazard events, a tornado hit

raises applications less in small counties than in larger counties, and policies that lower the cost of participating or shift the work to the state are associated with a smaller gap.

3. Data and Empirical Approach

3.1 Unit of Analysis and Sample

The unit of observation is a county designated in a major disaster declaration (declaration type DR) whose incident type is a tornado or severe storm and whose incident began between 1999 and 2021, taken from FEMA's Disaster Declarations Summaries (Federal Emergency Management Agency 2026). The end date leaves at least three years to observe applications and obligations through 2024. The sample contains 12,081 county–declarations covering 2,767 counties and 736 declarations in 49 states (Figure 1a). Tornado and severe-storm declarations are frequent across the central and eastern United States, so most counties in those regions appear several times.

3.2 Outcomes

Outcomes come from two OpenFEMA files (Federal Emergency Management Agency 2026). The Hazard Mitigation Assistance Projects file records funded and unfunded projects with their program, disaster number, project type, subrecipient, and federal share. The HMA Subapplications file records HMGP subapplications with their disaster number, benefiting counties, applicant name and type, requested amount, benefit–cost ratio, and status, including approved, withdrawn, denied, and pending requests. Projects that list several counties are credited to every listed county; statewide projects are kept separate and used in a robustness check.

The primary outcomes are binary indicators, for each county–declaration, of (i) any HMGP subapplication benefiting the county under that declaration, (ii) any funded HMGP project under that declaration, and (iii) a funded tornado safe room or wind retrofit under that declaration, identified from FEMA project-type codes. We also use the number of subapplications; a safe room or retrofit obligated within three or five years of the incident, regardless of disaster number; and, among county–declarations with at least one decided subapplication, the shares approved,

withdrawn, and denied. Applicant names identify subapplications and projects filed by the county government itself. Across county–declarations, the indicator for any approved subapplication and the indicator for any funded project have a correlation of 0.92.

These outcomes measure access to funding. They do not measure whether projects were completed or what losses they avoided.

3.3 County Size, Local Government Structure, Hazard Exposure, and Controls

Small counties are the bottom third of U.S. counties by baseline population, which corresponds to at most 15,136 residents; 34.7 percent of county–declarations fall in this group (Figure 1b). Because a binary cutoff is arbitrary, we also estimate the gap across population deciles and with continuous log population. Population is a summary measure: it bundles the number of potential applicants, staff, tax base, rurality, and exposure. The analysis therefore treats population as the dimension along which the gap is described and uses separate measures to examine which of these attributes are associated with it.

Hazard exposure is measured from the SPC tornado database (NOAA Storm Prediction Center 2025). Each track is represented as a straight path from its start to its end point with the reported path width, extended by 500 meters on each side and intersected with 2020 county boundaries and block-group population centroids (Appendix S4). A county–declaration is classified as hit if, in the calendar year in which the incident began, a tornado path crossed the population centroid of at least one block group in the county; 11.1 percent of county–declarations meet this definition. We also record the number of residents in tornado paths, track length within the county, the maximum EF rating, and casualties. Because EF ratings are assigned from observed damage (Doswell, Brooks, and Dotzek 2009), exposure is defined from geometry and population rather than from ratings or damage reports. Reported property damage from NOAA's Storm Events database enters as a separate control, in levels and per resident.

County characteristics are median household income, the poverty rate, and the non-White population share from the 2009 American Community Survey, and nonmetropolitan status. The 2017 Census of Governments provides the number of municipalities and townships, special districts,

and school districts in each county, an indicator for whether the county government lists a website, and county-government tax revenue and spending on financial administration and central staff per resident (U.S. Census Bureau 2019). The counts of local governments describe the institutional structure of each county, that is, how many entities could submit a subapplication. The finance variables are coarse proxies for fiscal and administrative resources; none of the available measures captures technical expertise directly.

Table 1 summarizes the sample. Small counties filed HMGP subapplications in 14.6 percent of their county–declarations, compared with 26.3 percent for larger counties, and obtained funded projects in 15.1 and 27.2 percent of cases, respectively. Averaged across counties, small counties contain fewer municipalities and townships (8.4 versus 14.0), special districts (6.9 versus 15.1), and school districts (2.2 versus 6.0); Table 1 reports the corresponding means across county–declarations. Small counties were also less often hit by a tornado crossing a populated place.

3.4 Empirical Specification

The baseline model compares counties designated in the same declaration:

$$Y_{cd} = \beta\, Small_c + \mathbf{X}'_{cd}\boldsymbol{\gamma} + \mathbf{Z}'_c\boldsymbol{\delta} + \alpha_d + \varepsilon_{cd}, \tag{1}$$

where Y_{cd} is an outcome for county c under declaration d, $Small_c$ indicates the bottom population third, $\mathbf{X}_{cd}$ contains the hazard measures for the incident year (populated-place hit, log residents in paths, track length, maximum EF rating, log casualties, log damage, and log damage per resident), $\mathbf{Z}_c$ contains county characteristics, and α_d is a declaration fixed effect.

The declaration fixed effect is the core of the design. Counties designated in the same declaration share the incident, the size of the HMGP allocation, the state's administrative plan and priorities, the year, and any political considerations behind the declaration itself (Garrett and Sobel 2003; Reeves 2011). Comparing counties within a declaration therefore removes the differences in disaster severity, state administration, and available funding that confound comparisons across disasters. We estimate Equation (1) as a linear probability model, add controls in steps, and finally add the log

counts of local governments to separate the part of the gap associated with the number of potential applicants. The coefficient β measures how much less likely a small county is to obtain an outcome than a larger county in the same declaration with similar hazard exposure and characteristics.

To ask whether small counties respond differently to hazard events, we use the fact that most counties appear in several declarations, some of which brought a tornado through populated places and some of which did not:

$$Y_{cd} = \theta\, H_{cd} + \lambda\, (H_{cd} \times Small_c) + \mu_c + \alpha_d + \varepsilon_{cd}, \tag{2}$$

where H_{cd} is the populated-place hit indicator (or log residents in tornado paths), μ_c is a county fixed effect, and α_d a declaration fixed effect. The county effect absorbs persistent differences in need, capacity, and political orientation, so θ is the response of larger counties to a hit and $\theta + \lambda$ the response of small counties. Damage controls are omitted from Equation (2) because damage is the channel through which a hit would raise demand for mitigation.

Finally, we use the management-cost provision of the DRRA to ask whether lowering the cost of administering a subaward narrowed the gap:

$$Y_{cd} = \pi\, (Small_c \times Post_d) + \mathbf{X}'_{cd}\boldsymbol{\gamma} + \mu_c + \alpha_d + \varepsilon_{cd}, \tag{3}$$

for declarations made between 2010 and 2021, where $Post_d$ indicates declarations made on or after August 1, 2017. Because the provision applies from a single date to all declarations, the design does not involve staggered adoption, and a two-way fixed-effects estimator does not suffer from the weighting problems discussed for staggered designs (Roth et al. 2023). We report an event-study version with small-county interactions by declaration year and a placebo cutoff in the pre-reform period.

3.5 What the Design Identifies

Equation (1) describes how participation varies with county size; it does not estimate a causal effect of population or of administrative capacity. Three features make the comparison informative nonetheless. The declaration fixed effect removes differences in incident severity, state administration, and HMGP allocations. The hazard controls use track geometry and population rather than damage-based ratings, so a small county is compared with a larger county facing a similar physical hazard. And the application records separate the decision to apply from the state's and FEMA's decisions on submitted requests. The decomposition that adds counts of local governments is also descriptive: it shows how much of the size gap is statistically accounted for by the number of potential applicants, not how participation would change if that number changed. Equation (2) adds variation within counties, but hit status within a declaration is not randomly assigned, and the reform in Equation (3) coincided with other changes in federal mitigation policy. We therefore interpret these estimates as checks on whether the patterns are consistent with a capacity explanation, not as causal effects of capacity.

Standard errors are clustered by county, the level at which the population classification is assigned (Abadie et al. 2023; Cameron and Miller 2015); we also report state-clustered errors (47 clusters). Declarations with a single designated county drop out as singletons, leaving 12,038 observations in the declaration fixed-effects models. Estimation uses PyFixest (Fischer and Schär 2024).

4. Results

4.1 How Large Is the Small-County Gap?

Figure 2a plots raw participation by population decile. The share of county–declarations with any HMGP subapplication rises from 14.3 percent in the smallest decile (median population 3,426) to 43.2 percent in the largest (median 413,325), and the share with a funded project follows almost the same path. The two lines are close because most decided subapplications are approved, so

the differences between counties emerge before a request reaches the state or FEMA. The rise is steepest in the upper deciles, but participation also increases, with small reversals, across the lower half of the distribution.

Table 2 reports Equation (1) with controls added across columns. Comparing counties within the same state and year (column 1), small counties are 11.4 percentage points less likely to obtain any funded HMGP project (Panel B). Replacing state-by-year effects with declaration fixed effects (column 2) barely changes the estimate, and adding the hazard and damage controls (column 3) reduces it only to 10.5 points. The gap is therefore not explained by small counties experiencing weaker storms. Adding income, poverty, race, and nonmetropolitan status reduces the gap to 8.8 percentage points (column 4), or 32 percent of the larger-county mean of 27.1 percent. The gap in filing any subapplication (Panel A) is of the same size, 8.9 percentage points on a mean of 26.1 percent, an early sign that the shortfall occurs before any decision on a request is made.

The gap is proportionally larger for investments aimed at tornado risk. Small counties are 3.1 percentage points less likely to obtain a safe room or wind retrofit under the declaration (Panel C, column 4), against a larger-county mean of 6.3 percent, so their probability is about half that of larger counties. Over a three-year window that includes projects funded under later declarations, the gap is 8.1 percentage points on a mean of 14.6 percent (Table 5).

The gap is a gradient rather than a threshold (Figure 2b). Relative to the largest population decile, and within declarations with full controls, the probability of a funded project is 35.5 percentage points lower in the smallest decile, 23.1 points lower in the fifth, and 11.5 points lower in the ninth (Appendix Table S2); each decile lies below the one above it. In continuous form, a one-log-point increase in population, roughly a 2.7-fold difference, is associated with a 7.8 percentage point higher probability of a funded project (Table 5). Differences in participation thus arise across the whole range of county sizes and are not confined to a few very small counties.

4.2 How Much Reflects the Number of Potential Applicants?

A county's claim on an HMGP allocation is made by the entities located in it, and small counties have fewer of them. Column (5) of Table 2 adds the log numbers of municipalities and townships,

special districts, and school districts. The gap in funded projects falls from 8.8 to 5.0 percentage points, and the gap in subapplications from 8.9 to 5.4 points. About two-fifths of the small-county gap is thus statistically accounted for by the number of entities that could apply. The count of school districts has the largest coefficient of the three (Appendix Table S1), consistent with schools serving as sites for community safe rooms. For safe rooms and wind retrofits, local-government counts account for a smaller share of the gap (from 3.1 to 2.4 percentage points), leaving a gap equal to 38 percent of the larger-county mean. Both residual gaps are precisely estimated, with standard errors of at most one percentage point.

This component reflects the institutional structure of local government in small counties rather than the behavior of any single government. Because the counts are not assigned independently of county size, the decomposition describes how the gap is distributed across this channel; it does not show what would happen if a county gained or lost local governments.

4.3 Does the Gap Arise at Application or at Approval?

Table 3 and Figure 3 follow requests through the grant process, holding constant the column (5) controls, including local-government counts. Small counties are 5.4 percentage points less likely to file any subapplication and file about 4.7 percent fewer. Among county–declarations with at least one decided subapplication (2,515 observations), the approval rate of small counties differs from that of larger counties by −0.5 percentage points (95 percent confidence interval from −4.2 to 3.2) on a mean of 86.6 percent, and the withdrawal rate by −0.9 points (−4.4 to 2.7). The denial rate is 1.4 percentage points higher ($p = 0.03$), from a low base of 1.7 percent; this difference is statistically detectable but too small to account for a meaningful part of the gap. At the subapplication level, small-county requests are smaller and report lower benefit–cost ratios, but neither difference is statistically distinguishable from zero once local-government counts are controlled (Table 3, note). The evidence supports H2: the shortfall occurs at the decision to apply, not in the review of submitted requests. This comparison of decision rates relies mainly on recent records, because few older closed or denied subapplications carry the county information needed for linkage (Section **??**).

The county government provides the cleanest check on the role of institutional structure, because every county has exactly one. With the full set of controls, small county governments are 2.4 percentage points less likely to file an HMGP subapplication under a declaration, on a mean of 11.1 percent (column 3), and 2.4 points less likely to receive a project, on a mean of 11.4 percent (column 7). A gap of about one-fifth for a single, comparable applicant cannot be attributed to the number of entities, which supports H3.

4.4 Why Do Small Counties Apply Less?

Table 4 assembles four tests of explanations for the application gap.

Response to hazard events. Panel A estimates Equation (2). In larger counties, a declaration in which a tornado crossed a populated place raises the number of subapplications by 4.3 percent ($p = 0.02$) and the probability that the county government applies by 2.4 percentage points ($p = 0.045$), relative to other declarations designating the same county. In small counties the corresponding responses are -2.9 percent and -1.8 points, and the differences between small and larger counties are -7.2 percent ($p = 0.046$) and -4.2 points ($p = 0.06$). Figure 4 displays these responses. The continuous measure, log residents in tornado paths, gives the same pattern (0.35 points per log unit for larger county governments, $p = 0.02$; interaction -0.60, $p = 0.05$). For the broader outcomes, any subapplication and any funded project, the interactions have the same sign but are imprecise ($p = 0.23$ and $p = 0.16$). Because the county fixed effects remove persistent differences between places, this pattern is difficult to reconcile with the view that small counties simply need less mitigation: larger counties respond to a damaging event with more applications, while small counties show no comparable response. With p-values near conventional thresholds and several outcomes examined, this evidence for H4 is suggestive rather than conclusive.

Fiscal and administrative resources. Panel B considers fiscal and administrative capacity measured per resident. Small county governments raise more tax revenue per resident and spend more per resident on financial administration and central staff than larger ones, consistent with the fixed cost of operating a government being spread over fewer residents. Controlling for these measures leaves the gap in subapplications at 5.4 percentage points, and the gap is not larger

where per-resident administrative spending is low (interaction −0.5 points, standard error 1.6). Holding population constant, absolute administrative spending and tax revenue also do not predict applications. The ability to fund the 25 percent match, at least as captured by these fiscal measures, does not explain the gap. These measures record spending, not the specialized expertise that applications require, so this result does not rule out a role for capacity more broadly.

State delivery. Panel C asks whether states that file applications on behalf of localities have smaller gaps. Using the share of each state's HMGP subapplications filed by state agencies before 2009 (mean 18.5 percent), a one-standard-deviation higher state share is associated with a 2.6 percentage point smaller gap in subapplications ($p = 0.04$) and a 3.0 point smaller gap in funded projects ($p = 0.02$) in 2009–2021. A comparable measure for regional planning and development bodies, which file few subapplications in any state, shows no association. The state-share result is a cross-state association, and states that file more applications may differ in other ways, but it is consistent with survey evidence that state assistance shapes local access (Smith and Vila 2020).

Reimbursement of management costs. Panel D and Figure 5 test whether reimbursing subrecipients' management costs narrowed the gap. In raw terms, the gap in subapplications was 16.2 percentage points for declarations made from 2010 to July 2017 and 12.4 points afterward. Within counties and declarations, however, the change is −1.9 percentage points (standard error 3.6) for subapplications and −1.7 points (3.6) for funded projects, and the placebo comparison at August 2014 is likewise null. The event-study coefficients fluctuate around zero before and after the reform without a break (smallest $p = 0.14$). The estimates are imprecise: the upper end of the 95 percent confidence interval for subapplications corresponds to a narrowing of about 5 percentage points, so a modest effect cannot be excluded, but a narrowing of the size of the full baseline gap is not supported. A smaller exploratory comparison around FEMA's pre-calculated benefits for residential safe rooms, whose start date we cannot establish precisely, also shows no differential change for small counties (Appendix S3).

4.5 Robustness and Qualifications

Table 5 reports the gap in funded projects under alternative choices. Clustering by state increases the standard error only modestly for the main outcome. Restricting the sample to county–declarations in which a tornado crossed a populated place, where demand for wind mitigation should be highest, enlarges the gap to 14.3 percentage points on a mean of 33.8 percent and to 9.2 points for safe rooms and retrofits on a mean of 14.5 percent. Limiting the sample to declarations that included Individual Assistance leaves the gap in any funded project unchanged (−8.8 points) and somewhat enlarges the gap in safe rooms and retrofits (−4.3 points). Dropping each state in turn yields gaps between 8.4 and 9.4 percentage points for any funded project and between 2.7 and 3.5 points for safe rooms and retrofits. The gap over a five-year window is 10.5 points on a mean of 19.7 percent.

Two checks qualify the results. When statewide HMGP projects under the same declaration are credited to every designated county, the gap in any project nearly disappears (−0.9 points). This outcome is close to saturated, however, because about 89 percent of larger counties' county–declarations, and 86 percent of small counties', are covered by some statewide project, mostly planning and similar activities. For safe rooms and retrofits the gap remains (−2.8 points on a mean of 9.9 percent), so state-run projects offset only part of the shortfall in local investment. Among the 361 county–declarations under declarations whose incident type is recorded as a tornado rather than a severe storm, the estimates are small and imprecise (−1.5 points, standard error 4.0); this subsample is too small to yield informative estimates.

5. Discussion

5.1 Geographic Access to Intergovernmental Investment

After tornado and severe-storm disasters, small counties obtain HMGP projects much less often than larger counties designated in the same declaration, and the shortfall arises because they apply less often, not because their applications fare worse. A grant program whose funding is triggered by hazards can therefore still distribute investment unevenly across places that face similar hazards,

because the step that turns an allocation into a project depends on local action. Unequal exposure to hazards is not the source of this disparity. It arises in how the grant is delivered. Post-disaster assistance has similarly been found to widen rather than narrow differences in wealth (Howell and Elliott 2019).

Several explanations can be set aside or bounded. Lower need is unlikely to explain the full gap: the comparisons hold the declaration fixed and control for tornado exposure and damage, and the gap is larger, not smaller, among county–declarations in which a tornado crossed a populated place. Federal or state screening of applications does not explain it, since approval and withdrawal rates are similar once a request is filed. Income, poverty, race, and nonmetropolitan status account for part of the raw difference but leave most of it unexplained, and per-resident fiscal measures add nothing.

A substantial part of the gap is associated with the institutional structure of local government. Small counties have fewer cities, school districts, and special districts that could apply, so they make fewer claims on a declaration's allocation even if each entity behaves like its counterparts elsewhere. This component is a feature of how local government is organized in small counties rather than a failure of any single government. The remaining gap, visible for county governments themselves and in the weaker response to tornado hits, is consistent with the capacity account developed in public administration and planning research (Collins and Gerber 2006; Hall 2008; Lyles, Berke, and Smith 2014; Smith et al. 2025): preparing an HMGP application requires expertise and staff time whose cost does not fall in proportion to population. The data do not measure these costs, however, and the residual could also reflect other attributes that vary with county size.

The null result for management-cost reimbursement is informative about timing. If the binding constraint were the ongoing cost of administering a subaward, reimbursing up to 5 percent of it should have helped small counties disproportionately. We find no detectable change, although the estimate is imprecise. This is consistent with a constraint that operates before any award exists, in scoping a project, assembling cost estimates, and completing a benefit–cost analysis, for which reimbursement after an award offers little help. The association with state delivery points in the

same direction: where the state prepared more of the applications, the gap is smaller. Neither piece of evidence is causal, but together they suggest that up-front technical assistance is a more promising policy margin than reimbursement after an award.

5.2 Relation to Previous Findings

Seong, Losey, and Gu (2022) find that rural HMGP recipients receive less per recipient. Our results suggest that an important margin may be whether there is a recipient at all, and that this margin varies continuously with population rather than with a rural classification. Miao, Davlasheridze, and Reilly (2025) report that aid programs administered through state and local governments favor better-off counties. Our evidence locates one channel for that pattern in the decision to apply, and it shows that, for HMGP after tornado and storm declarations, participation is more closely tied to county size than to income: conditional on size and the other controls, median income is only weakly associated with participation and the poverty rate not at all (Table 2, column 4; coefficients in Appendix Table S1). Frimpong, Reilly, and Niemeier (2023) find that mitigation grants do not raise participation in the Community Rating System. Our findings concern the opposite direction, from local conditions to grant participation, and together the two studies describe a system in which local administrative effort and federal mitigation funding are only loosely connected.

5.3 Implications for Grant Design

The findings have three implications for how place-based mitigation grants are designed and evaluated. First, evaluations that compare outcomes across counties with and without mitigation projects should recognize that receiving a project depends on factors associated with county size, which are likely correlated with other determinants of resilience. Second, programs that distribute funds through local applications will tend to direct investment toward larger jurisdictions even when hazard exposure is similar, so equity goals stated in terms of need or vulnerability may not be met without attention to the application stage. Third, the gap is proportionally larger for tornado-specific projects, and small counties in the central and southeastern United States include many places where tornado risk and mobile-home housing are common (Ashley 2007; Sutter and

Simmons 2010). Options consistent with the evidence include applications prepared by the state or by regional bodies on behalf of small jurisdictions, pre-application technical assistance, and simplified cost-effectiveness requirements for standard project types. The present analysis does not evaluate the effects of these options.

6. Conclusion

Federal mitigation grants are meant to turn the aftermath of a disaster into an opportunity to reduce future risk, but in HMGP that opportunity becomes a project only when a local government applies. Across two decades of tornado and severe-storm declarations, the smallest third of counties obtained HMGP projects in roughly one fewer case out of every ten county–declarations than larger counties designated in the same declaration, and they were about half as likely to obtain the safe rooms and wind retrofits that address tornado risk directly. The gap opens before any request is reviewed.

For regional economics, the main lesson concerns the delivery of intergovernmental transfers. When federal funds for local public investment must be claimed through applications, the distribution of investment across places reflects how local government is organized and the capacity of individual governments to apply, as well as hazard and need. Part of the small-county gap is associated with the smaller number of local entities in small counties; the rest appears for county governments themselves and in their weaker response to tornado hits. Reimbursing management costs after an award did not detectably change this pattern, while state preparation of applications is associated with a smaller gap. These results are descriptive, and identifying the causal effect of local capacity on grant participation, for example through variation in technical assistance across states or over time, remains a task for future research. The evidence nonetheless indicates that reducing geographic disparities in mitigation investment requires attention to how applications are prepared, and not only to how funds are allocated.

Statements and Declarations

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Conflict of Interest

The author declares no conflicts of interest relevant to this study.

Data and Code Availability

All data used in this study are publicly available: FEMA's OpenFEMA Hazard Mitigation Assistance Projects, HMA Subapplications, and Disaster Declarations Summaries; the NOAA Storm Prediction Center tornado database; NOAA Storm Events; and the U.S. Census Bureau's 2020 block-group centers of population, 2009 American Community Survey, and 2017 Census of Governments. Code that reproduces all tables and figures from these sources will be deposited in a public repository upon acceptance.

Declaration of Generative AI Use

During the preparation of this work, the author used Claude (Anthropic) solely for grammar and language editing. The author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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Table 1. Summary statistics, county–declaration observations, 1999–2021

		All		Mean by county size	
Variable	*N*	Mean	SD	Small	Larger
Outcomes (per county-declaration)					
Any HMGP subapplication filed	12,081	0.222	0.416	0.146	0.263
Number of HMGP subapplications	12,081	0.5	2.1	0.2	0.6
Any funded HMGP project	12,081	0.230	0.421	0.151	0.272
Funded safe room or wind retrofit	12,081	0.050	0.219	0.025	0.064
Safe room or wind retrofit obligated within 3 years	12,081	0.122	0.327	0.075	0.146
Federal HMGP dollars per resident	12,081	3.8	22.8	5.3	3.0
Hazard exposure					
Tornado path crossed a populated place	12,081	0.111	0.314	0.046	0.146
Residents in tornado paths	12,081	586.8	4,037.8	60.3	866.1
Tornado track length in county (km)	12,081	5.1	12.9	4.3	5.5
Reported property damage ($ million)	12,081	2.4	42.2	0.4	3.5
County characteristics					
Population (thousands)	12,081	79.4	265.0	8.3	117.2
Median household income ($ thousand)	12,081	41.9	10.9	38.0	44.0
Poverty rate	12,081	0.157	0.065	0.162	0.154
Non-White population share	12,081	0.131	0.149	0.111	0.142
Nonmetropolitan	12,081	0.661	0.474	0.885	0.542
Municipalities and townships	11,972	12.5	12.8	9.8	13.9
Special districts	11,972	12.1	18.3	7.5	14.6
School districts	11,972	4.8	6.9	2.3	6.2
County government lists a website	11,972	0.637	0.481	0.485	0.718
Any prior HMA project	12,081	0.720	0.449	0.656	0.753

Notes: One observation is a county designated in a major disaster declaration (DR) for a tornado or severe-storm incident that began in 1999–2021 (12,081 county–declarations; 2,767 counties; 736 declarations). Small counties are the bottom third of U.S. counties by baseline population (at most 15,136 residents). HMGP outcomes are linked to the specific declaration; subapplications include withdrawn, denied, and pending records. Tornado exposure is constructed from Storm Prediction Center tracks and 2020 block-group population centroids; damage is from NOAA Storm Events; demographics from the 2009 American Community Survey; local-government counts from the 2017 Census of Governments.

Table 2. The small-county gap in HMGP participation after tornado and severe-storm declarations

	(1)	(2)	(3)	(4)	(5)
Panel A. Any HMGP subapplication filed					
Small county	−11.27***	−11.19***	−10.43***	−8.85***	−5.37***
	(0.88)	(0.88)	(0.88)	(0.93)	(0.98)
Mean, larger counties	26.2	26.1	26.1	26.1	26.1
Panel B. Any funded HMGP project					
Small county	−11.43***	−11.25***	−10.51***	−8.79***	−4.99***
	(0.90)	(0.89)	(0.89)	(0.95)	(0.99)
Mean, larger counties	27.1	27.1	27.1	27.1	27.1
Panel C. Funded safe room or wind retrofit					
Small county	−3.97***	−3.85***	−3.45***	−3.14***	−2.38***
	(0.53)	(0.53)	(0.52)	(0.53)	(0.54)
Mean, larger counties	6.4	6.3	6.3	6.3	6.3
Observations	12,057	12,038	12,038	12,038	12,038
Counties (clusters)	2,763	2,762	2,762	2,762	2,762
Fixed effects	State×year	Decl.	Decl.	Decl.	Decl.
Hazard exposure and damage			√	√	√
Income, poverty, race, nonmetro				√	√
Counts of local governments					√

Notes: Linear probability models estimated by OLS. Coefficients and means are multiplied by 100 and read as percentage points. "Small county" is an indicator for the bottom third of counties by population. Standard errors clustered by county in parentheses. Declaration fixed effects compare counties designated in the same disaster declaration, which share the same HMGP allocation, state administration, and incident. Hazard controls: populated-place tornado hit, log residents in tornado paths, track length, maximum EF rating, log casualties, log reported damage, and log damage per resident. Declarations with a single designated county drop out of columns (2)–(5) as singletons. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 3. Where the gap arises: applications versus decisions

	Application margin			Share of decided subapplications			Award
	Applied	Log(1+ number)	County gov. applied	Approved	Withdrawn	Denied	County gov. funded
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Small county	−5.37***	−4.68***	−2.35***	−0.52	−0.87	1.40**	−2.38***
	(0.98)	(1.02)	(0.73)	(1.90)	(1.82)	(0.65)	(0.74)
Mean, larger counties	26.1	25.9	11.1	86.6	11.7	1.7	11.4
Observations	12,038	12,038	12,038	2,515	2,515	2,515	12,038
Counties (clusters)	2,762	2,762	2,762	1,459	1,459	1,459	2,762

Notes: All columns use the column (5) specification of Table 2 (declaration fixed effects; hazard, demographic, and local-government-count controls). Coefficients multiplied by 100. Columns (1)–(3) and (7) use all county–declarations; "county gov." restricts to subapplications or projects whose applicant is the county government itself, of which every county has exactly one. Columns (4)–(6) restrict to county–declarations with at least one decided subapplication and measure the share approved, withdrawn, or denied. Standard errors clustered by county. At the subapplication level (same controls), small-county subapplications request 0.102 log points less funding (SE 0.094) and report benefit–cost ratios 0.069 log points lower (SE 0.063). $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 4. Testing explanations for the small-county application gap

	Applied (1)	Log(1+ number) (2)	County gov. applied (3)	Funded (4)
Panel A. Same county, different declarations (county and declaration fixed effects)				
Tornado hit a populated place	2.40	4.32**	2.41**	2.35
	(1.64)	(1.80)	(1.20)	(1.61)
× small county	−4.08	−7.23**	−4.19*	−4.75
	(3.38)	(3.63)	(2.22)	(3.35)
Log residents in tornado paths	0.33	0.60**	0.35**	0.33
	(0.21)	(0.24)	(0.15)	(0.21)
× small county	−0.62	−1.04**	−0.60**	−0.72
	(0.47)	(0.50)	(0.31)	(0.46)
Observations	11,491	11,491	11,491	11,491
Panel B. Fiscal and administrative capacity (declaration fixed effects, full controls)				
Small county, controlling for taxes and administrative spending per resident	−5.38***	−4.71***	−2.52***	−4.98***
	(0.98)	(1.02)	(0.73)	(0.99)
Small × low administrative spending	−0.47	0.03	1.40	−0.31
	(1.62)	(1.58)	(1.21)	(1.66)
Observations	12,038	12,038	12,038	12,038
Panel C. State delivery (declarations 2009–2021; declaration fixed effects, full controls)				
Small × state-filed share (SD), pre-2009	2.63**	1.73	0.81	3.03**
	(1.28)	(1.27)	(0.93)	(1.28)
Small × regional-body share (SD), pre-2009	−0.95	−0.56	−0.43	−1.23
	(1.02)	(0.91)	(0.74)	(1.04)
Observations	5,805	5,805	5,805	5,805
Panel D. Management-cost reimbursement, DRRA §1215 (county and declaration fixed effects)				
Small × declared on/after 1 Aug 2017	−1.92	1.39	−0.80	−1.74
	(3.61)	(3.61)	(2.48)	(3.62)
Placebo: small × declared on/after 1 Aug 2014 (pre-2017 sample)	−4.14	−0.50	−1.46	−4.43
	(3.92)	(3.68)	(2.83)	(3.96)
Observations	2,898	2,898	2,898	2,898

Notes: Coefficients multiplied by 100. Outcomes: (1) any HMGP subapplication; (2) log one plus the number of subapplications; (3) a subapplication by the county government; (4) any funded HMGP project. Panel A omits damage controls because damage is the channel through which a hit operates. Panel B uses Census of Governments 2017 finance data; "low" is the bottom third of county-government administrative spending (financial administration and central staff) per resident. Panel C interacts the small-county indicator with the standardized share of each state's FY1989–2008 HMGP subapplications filed by state agencies or by regional planning and development bodies; the main effects are absorbed by declaration fixed effects. Panel D compares declarations made before and after 1 August 2017, the date from which DRRA §1215 reimbursement of subrecipient management costs applies; the placebo repeats the comparison at a false date within the pre-reform period. Standard errors clustered by county. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 5. Robustness of the small-county gap in funded projects

	Funded HMGP project	Funded safe room/retrofit	Safe room/ retrofit, 3 yrs	*N*
Baseline (Table 2, column 4)	−8.79*** (0.95)	−3.14*** (0.53)	−8.08*** (1.11)	12,038
Standard errors clustered by state	−8.79*** (0.98)	−3.14*** (0.82)	−8.08*** (2.47)	12,038
Hit-only sample (populated-place tornado hit)	−14.25*** (3.77)	−9.18*** (2.60)	−13.38*** (4.60)	1,203
Declarations with incident type "Tornado" only	−1.46 (4.05)	−2.88 (2.16)	−2.71 (3.38)	361
Declarations with Individual Assistance	−8.81*** (1.57)	−4.34*** (0.90)	−8.92*** (1.64)	3,647
Adding counts of local governments	−4.99*** (0.99)	−2.38*** (0.54)	−5.93*** (1.14)	12,038
Continuous log population (per log point)	7.78*** (0.51)	1.87*** (0.27)	5.06*** (0.52)	12,038
Credit statewide projects of the same declaration	−0.93*** (0.27) [88.6]	−2.79*** (0.50) [9.9]	–	12,038
Five-year window	–	–	−10.46*** (1.42) [19.7]	12,038
Leave one state out: range of estimates	[−9.37, −8.43]	[−3.53, −2.70]	[−8.95, −6.58]	–

Notes: Each cell reports the small-county coefficient (multiplied by 100) with its county-clustered standard error in parentheses, from the column (4) specification of Table 2 unless stated. Means for larger counties of the two alternative outcomes are in brackets. "Credit statewide projects" counts a county as served when the state ran any HMGP project under the same declaration on a statewide basis. The leave-one-state-out row reports the minimum and maximum of the estimates obtained by dropping each state in turn. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

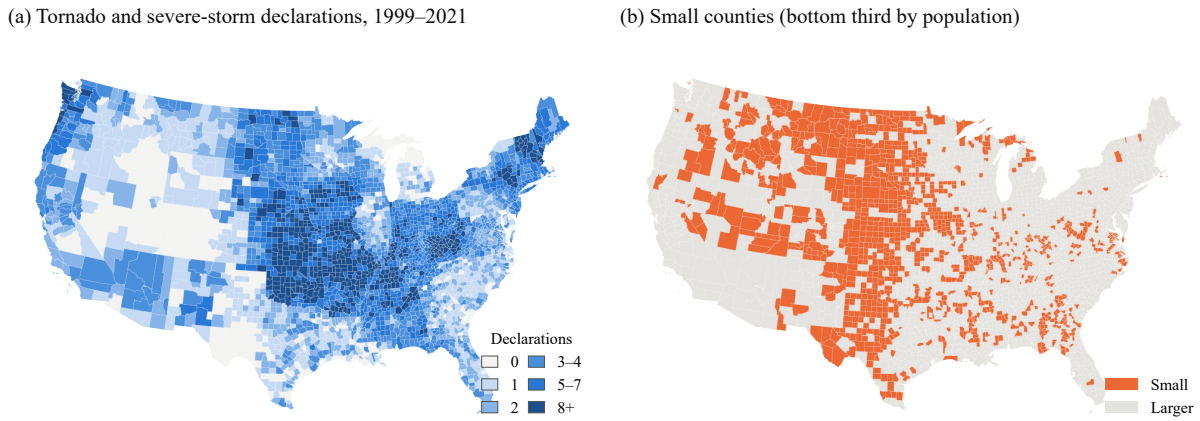


Figure 1. Study area. (a) Number of tornado and severe-storm major disaster declarations, 1999–2021, designating each county. (b) Small counties, defined as the bottom third of U.S. counties by baseline population. Alaska, Hawaii, and territories are not shown.

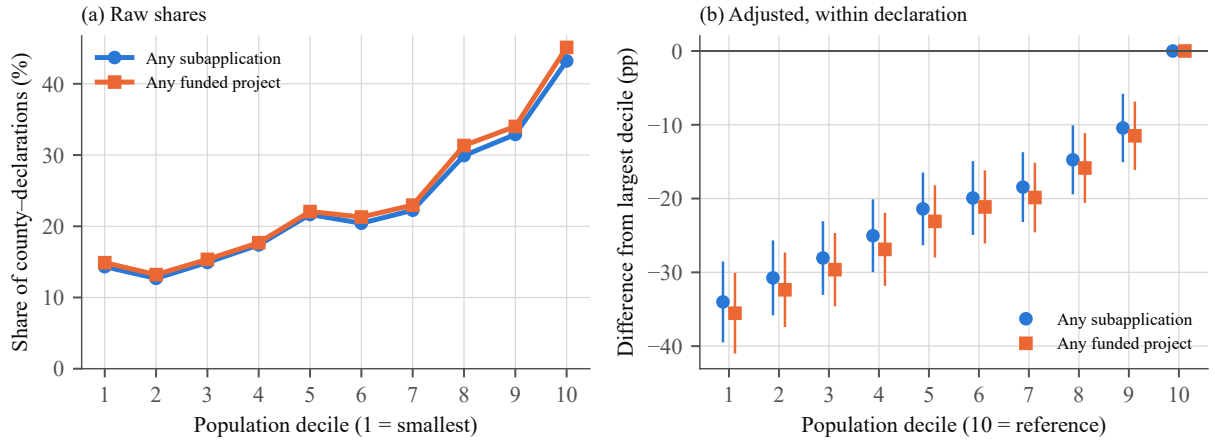


Figure 2. Participation in HMGP by county population decile. (a) Raw share of county–declarations with any HMGP subapplication and with any funded HMGP project under the declaration. (b) Differences from the largest decile, estimated with declaration fixed effects and hazard and demographic controls (Table 2, column 4); vertical bars are 95 percent confidence intervals with county-clustered standard errors.

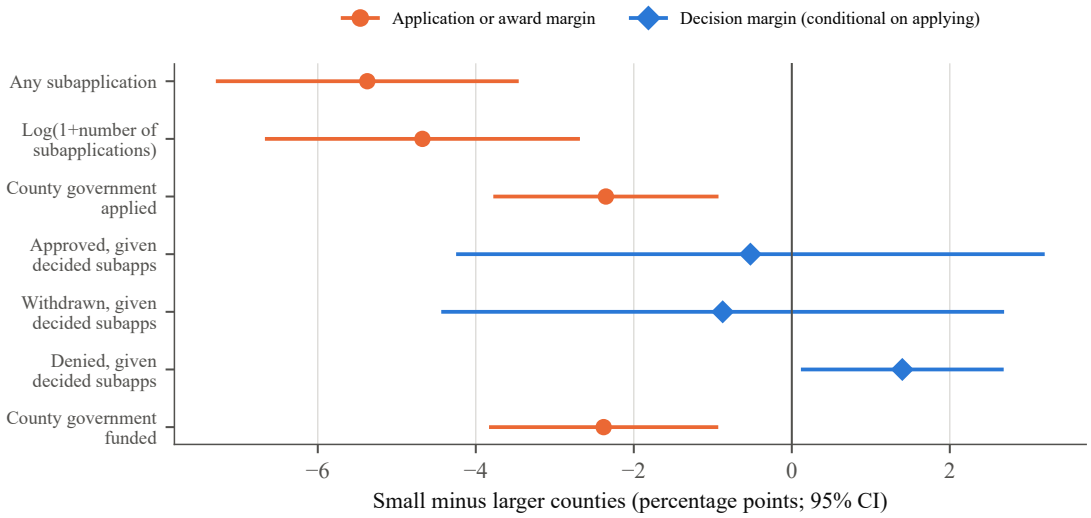


Figure 3. Where the small-county gap arises. Estimated differences between small and larger counties designated in the same declaration, with full controls including local-government counts (Table 3). Circles: application and award outcomes for all county–declarations. Diamonds: shares of decided subapplications approved, withdrawn, or denied, among county–declarations with at least one decided subapplication. Horizontal bars are 95 percent confidence intervals.

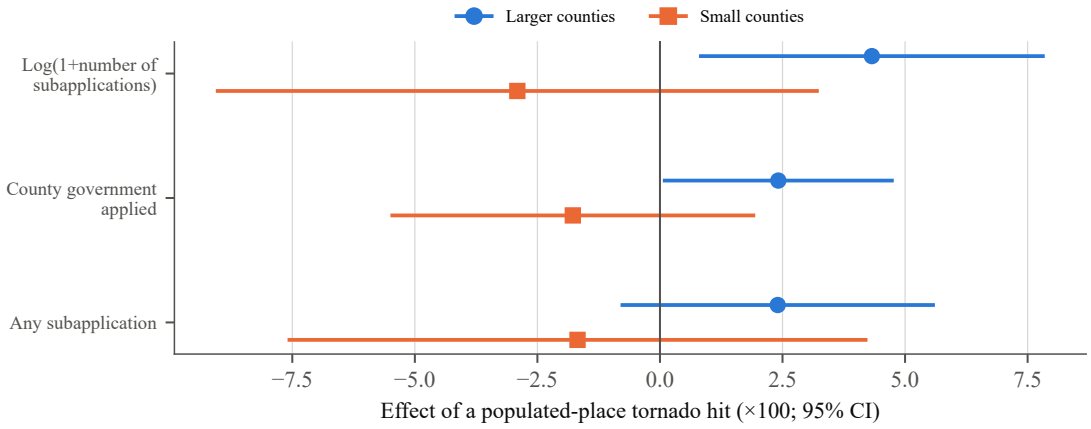


Figure 4. Within-county response to a tornado hit. Effect of a declaration in which a tornado crossed a populated place in the county, relative to other declarations designating the same county, estimated with county and declaration fixed effects (Equation 2). The small-county response is the sum of the main effect and the interaction, with its standard error computed from the full covariance matrix. Coefficients multiplied by 100; bars are 95 percent confidence intervals.

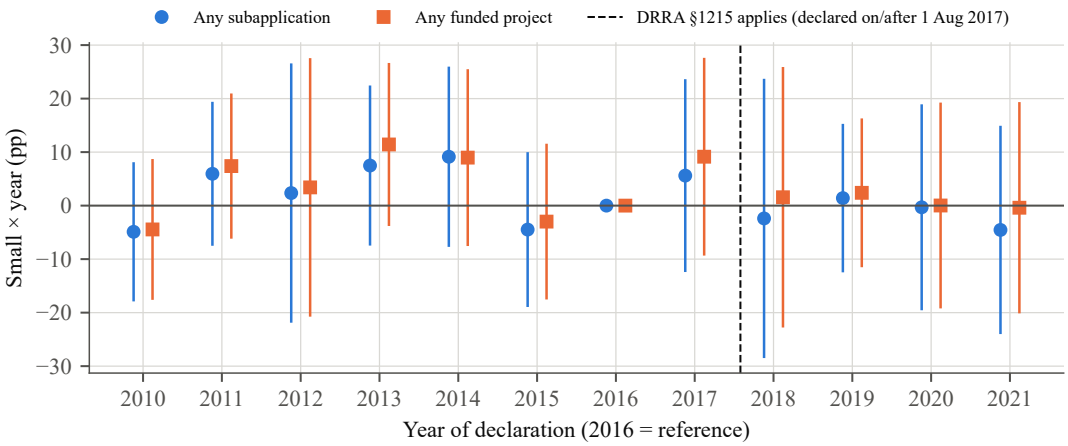


Figure 5. Small-county gap by year of declaration around the DRRA management-cost provision. Interactions of the small-county indicator with declaration-year indicators (2016 omitted), with county and declaration fixed effects and hazard controls, declarations of 2010–2021. The dashed line marks the date from which DRRA §1215 reimbursement of subrecipient management costs applies. Bars are 95 percent confidence intervals.

Supplementary Appendix

Appendix S1. Coefficients on all regressors

Table S1 reports every coefficient in the baseline specifications of Table 2. Conditional on population size and hazard exposure, log median household income is weakly and positively associated with participation, the poverty rate is not associated with it, the non-White population share is positively associated with it, and nonmetropolitan counties are about three percentage points less likely to participate. Among the local-government counts, the number of school districts has the largest coefficient.

Table S1. Coefficients on all regressors in the baseline specifications

	Any subapplication		Any funded project	
	(4)	(5)	(4)	(5)
Small county	−8.85*** (0.93)	−5.37*** (0.98)	−8.79*** (0.95)	−4.99*** (0.99)
Tornado path crossed a populated place	−20.20** (9.60)	−18.11* (9.40)	−23.01** (9.50)	−20.75** (9.21)
Log residents in tornado paths	2.98** (1.25)	2.61** (1.23)	3.37*** (1.24)	2.98** (1.20)
Track length in county (km)	−0.01 (0.04)	−0.02 (0.04)	−0.02 (0.04)	−0.03 (0.04)
Maximum EF rating (+1)	−0.23 (0.54)	−0.24 (0.53)	0.05 (0.55)	0.03 (0.54)
Log casualties	1.53** (0.60)	1.45** (0.59)	1.36** (0.60)	1.27** (0.59)
Log reported damage	0.06 (0.16)	−0.07 (0.16)	−0.04 (0.16)	−0.18 (0.16)
Log damage per resident	0.22 (0.55)	0.73 (0.55)	0.37 (0.55)	0.93* (0.55)
Log median household income	6.59* (3.92)	4.21 (3.85)	6.99* (3.98)	4.38 (3.89)
Poverty rate	−1.19 (13.63)	−3.73 (13.26)	−3.91 (13.72)	−6.70 (13.29)
Non-White share	9.80** (4.19)	7.85* (4.15)	10.59** (4.23)	8.51** (4.15)
Nonmetropolitan	−3.24*** (1.05)	−2.40** (1.05)	−3.47*** (1.08)	−2.57** (1.07)
Log municipalities and townships		2.98*** (0.91)		3.36*** (0.92)
Log special districts		1.38* (0.81)		1.38* (0.83)
Log school districts		4.66*** (1.15)		5.13*** (1.16)

Notes: Columns correspond to columns (4) and (5) of Table 2, with declaration fixed effects. Coefficients multiplied by 100; county-clustered standard errors in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Appendix S2. Population-decile gradient

Table S2 reports the coefficients plotted in Figure 2b.

Appendix S3. Exploratory test: pre-calculated benefits for residential safe rooms

FEMA's job aid for safe-room applications using pre-calculated benefits is dated February 19, 2015 (Federal Emergency Management Agency 2015); we were unable to establish the exact date from which the option was first available. As an exploratory check, we stacked two outcomes for each county–declaration of 2009–2021, an indicator for any safe-room or wind-retrofit subapplication and an indicator for any other subapplication, and estimated a triple-difference model with county-by-outcome and declaration-by-outcome fixed effects. The coefficient on small × declared after February 19, 2015 × safe-room outcome is 0.73 percentage points (county-clustered standard error 2.95; $p = 0.81$; 9,856 stacked observations). Because the treatment date is uncertain and the test has limited power, we do not interpret this estimate further.

Table S2. Population-decile gradient (relative to the largest decile)

Population decile	Any subapplication	Any funded project
1	−34.01 (2.80)	−35.54 (2.79)
2	−30.75 (2.59)	−32.36 (2.57)
3	−28.06 (2.54)	−29.64 (2.53)
4	−25.04 (2.52)	−26.89 (2.52)
5	−21.40 (2.51)	−23.09 (2.50)
6	−19.92 (2.55)	−21.13 (2.53)
7	−18.45 (2.41)	−19.86 (2.40)
8	−14.75 (2.38)	−15.86 (2.41)
9	−10.43 (2.37)	−11.50 (2.36)
10	0 (reference)	0 (reference)

Notes: Declaration fixed effects with hazard and demographic controls (Table 2, column 4). Coefficients multiplied by 100; county-clustered standard errors in parentheses.

Appendix S4. Data construction

Declarations. County designations are taken from FEMA's Disaster Declarations Summaries for declarations of type DR with incident type "Tornado" or "Severe Storm" and incident begin dates from 1999 to 2021. Statewide designations (county code 000) are excluded. The calendar year in which the incident began is used to align hazard measures.

HMGP projects and subapplications. Funded HMGP projects are linked to county–declarations through the disaster number and the project's county; projects listing several counties are credited to each listed county, and projects with no county (statewide) are used only in the statewide-credit robustness check. Safe rooms and wind retrofits are identified from FEMA project-type codes for community and individual safe rooms and for wind retrofits of structures. Subapplications are linked through the disaster number and the benefiting-county field, matched to county names within the subapplicant's state; void records are excluded. Of 32,607 linked county–subapplication records, 23,567 are approved, 4,199 withdrawn, 330 not approved, 224 subaward accepted, and 4,287 pending or submitted. Few closed or "not approved/denied" records carry benefiting counties, so decision rates rest mainly on the categories listed.

Applicant type. Applicant names were classified by text rules into county government, municipality or township, school district, special district or other, and state agency. A county-government applicant is one whose name identifies the county or parish government.

Tornado exposure. Each SPC tornado is represented by a straight segment from its start to end coordinates, buffered by half the reported path width on each side (minimum width 10 yards); tracks without an end point are represented by a disc with the reported swept area. Paths are intersected with 2020 county boundaries. A county–declaration is classified as hit if, in the incident year, a path buffered by an additional 500 meters contains the population centroid of at least one 2020 block group in the county. Residents in paths sum the 2020 population of such block groups.

Local governments and finances. Counts of municipalities and townships, special districts, and school districts (independent and dependent) come from the 2017 Census of Governments

Government Units Listing. County-government taxes (item codes beginning with T) and spending on financial administration and central staff (codes E23 and E29) come from the 2017 Individual Unit File.

State delivery measures. For each state, the share of HMGP subapplications for fiscal years up to 2008 filed by state agencies (names indicating a state department, division, or emergency management agency, or statewide benefit) and by regional planning or development bodies (names containing planning, regional, council of governments, or development district). These shares are measured before the 2009–2021 outcome period.